

SERVICE AND WIRING SHEET

W10674984A



⚠ WARNING

Electrical Shock Hazard
Disconnect power before servicing.
Replace all parts and panels before operating.
Failure to do so can result in death or electrical shock.

* Normal operating conditions are viewed when the air and temperature controls are at mid-setting, freezer section 0° to -5°F, and unit is cycling.

NOTE: Watt and pressure readings will vary and are influenced by the existing condition of the appliance, such as iced-up evaporator, condition of condenser, defrost cycle, pull-down time and customer use.

PERFORMANCE DATA *(NORMAL OPERATING CONDITIONS)			
AMB	WATTS	SYSTEM PRESSURE (PSIG)	
		HIGH SIDE	LOW SIDE
70°	120±20	120 ± 20	-6" TO 3#
90°	130±20	160 ± 20	-4" TO 3#
110°	140±20	220 ± 20	-2" TO 4#

SERVICE INFORMATION (W10674986 A)

1. COMPRESSOR SUCTION AND PROCESS STUBS MAY NOT BE INTERCHANGED UNLESS INDICATED BY **
2. REFRIGERANT CHARGE MUST BE APPLIED TO HIGH SIDE ONLY.
3. ICE MAKER AND WATER VALVE NOT ORIGINAL EQUIPMENT ON ALL MODELS.
4. CAUTION: ICE MAKER CYCLE MUST BE INITIATED ELECTRICALLY. DO NOT TRY TO MANUALLY START CYCLE.
5. SERVICE DEFROST BI-METALS 58°F OPEN.
6. THE PART NUMBER CAN BE FOUND ON THE COMPONENT. USE A REPLACEMENT PART OF SIMILAR PERFORMANCE.

GENERAL COMPONENT INFORMATION FOR ALL REFRIGERATOR/FREEZER MODELS BE SURE TO USE CORRECT REPLACEMENT PARTS

COMPONENT (IF APPLICABLE)	EMBRACO						OHMS RESISTANCE
	EM3Z	WATTS * 120V	EM3D	WATTS * 120V			
COMPRESSOR	W10445423		W10575980				
RELAY-STARTING, EMB	NOTE #6		NOTE #6				
OVERLOAD PROTECTOR-T.I.	NOTE #6		NOTE #6				
COVER-TERMINAL	-						
RUN WINDINGS	-						-
START WINDINGS	-						-
COMPRESSOR ELECTRICAL	-						
RUN CAPACITOR	NOTE #6		NOTE #6				-
ADAPTIVE DEFROST	NOTE #6		NOTE #6				
DEFROST HEATER	NOTE #6	350-480	NOTE #6	350-480			42-30
DEFROST BI-METAL	NOTE #6		NOTE #6				-
EVAPORATOR FAN	NOTE #6	1.5-2.5	NOTE #6	1.7-3.7			-
CONDENSER FAN	NOTE #6	3.1-5.1	NOTE #6	1.6-3.6			-

STEPS TO ENTER ELECTRONIC DEFROST CONTROL TEST MODE (IF APPLICABLE)

OPTION #1 STEPS (BI-METAL CLOSED):

- #1 - POWER OFF TO REFRIGERATOR FOR AT LEAST 30 SECONDS
- #2 - THERMOSTAT OFF
- #3 - POWER ON TO REFRIGERATOR

OPTION #2 STEPS (BI-METAL CLOSED):

- #1 - THERMOSTAT OFF 15 SECONDS
- #2 - THERMOSTAT ON 5 SECONDS
- (REPEAT STEPS 1 AND 2 TWO MORE TIMES)
- #3 - THERMOSTAT OFF

ENTER TEST MODE

IN 3-8 SECONDS, CONTROL WILL TURN ON DEFROST HEATER. DEFROST HEATER WILL TURN ON FOR 18 MINUTES OR UNTIL BI-METAL OPENS. TO TERMINATE TEST EARLY, REMOVE POWER FROM REFRIGERATOR.

NOTE: IF OPTION #2 DOES NOT WORK, TRY OPTION #1 BEFORE REPLACING CONTROL. ALWAYS CHECK CONNECTIONS BEFORE REPLACING CONTROL.

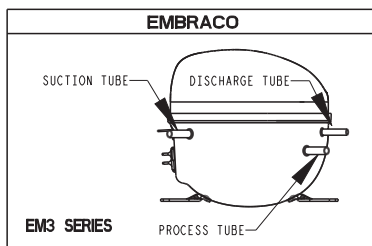
HELPFUL HINT: UPON ENTERING TEST MODE, A RELAY TURNS OFF THE COMPRESSOR AND TURNS ON THE DEFROST HEATER. LISTEN FOR THE RELAY TO CLICK. IF RELAY CLICKS ONCE UPON ENTERING TEST MODE, CHECK FOR DEFROST HEAT. IF RELAY CLICKS TWICE (ALLOW UP TO 30 SEC BETWEEN CLICKS), CHECK FOR BI-METAL OPEN.

⚠ WARNING

IF BI-METAL IS BY-PASSED FOR TESTING (IF APPLICABLE), DO NOT OVERHEAT EVAPORATOR AREA.

COMPRESSOR OPTIONS - REFER TO APPLICABLE DESIGN

(OIL COOLER NOT PRESENT ON ALL COMPRESSORS)



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NOTES:

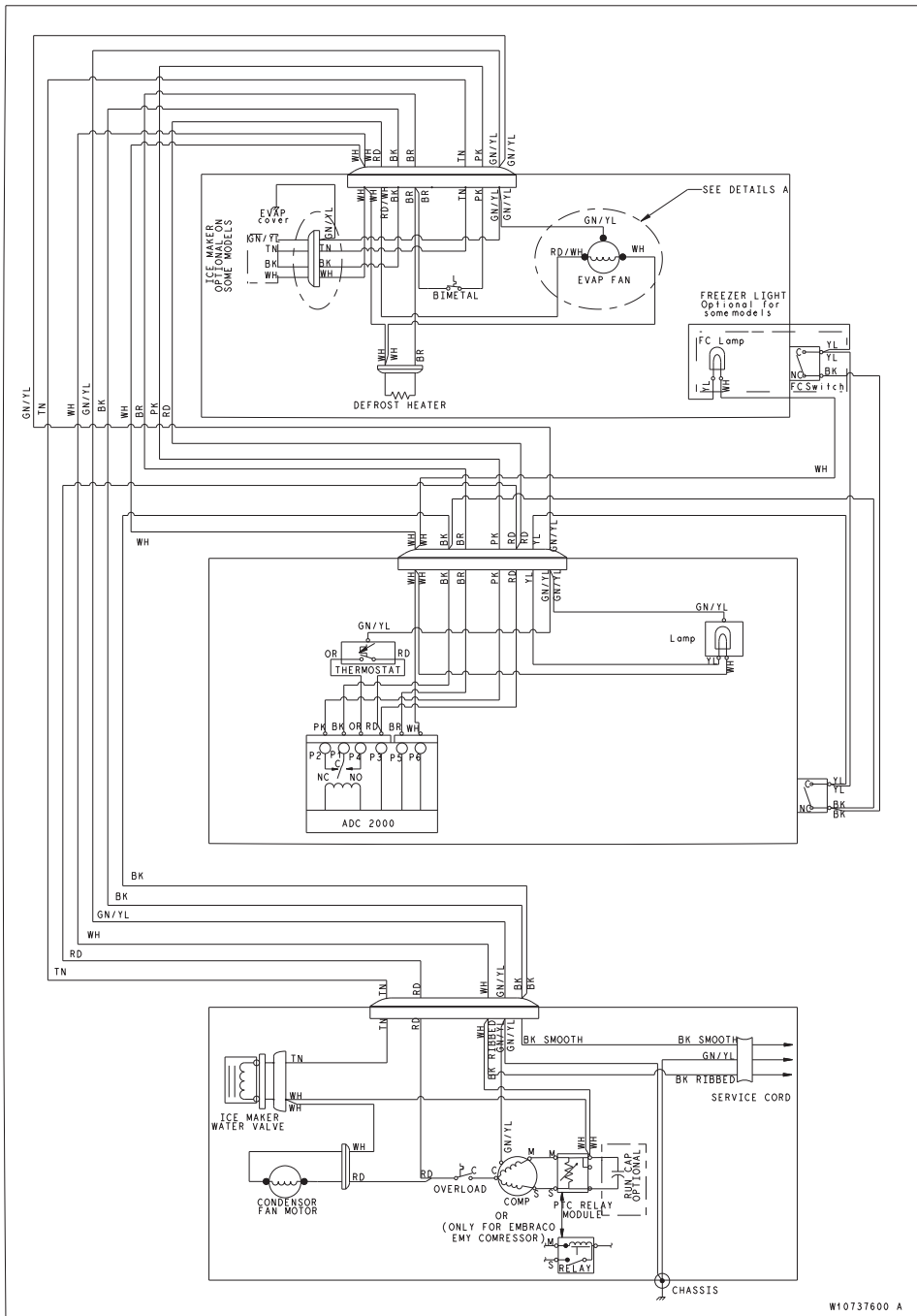
1. GROUND COMPRESSOR THROUGH CONDENSER.
2. 1M SOLENOID VALVE GROUNDED THROUGH MOUNTING.
3. EVAP COVER GROUNDED THROUGH HEAT SHIELD.



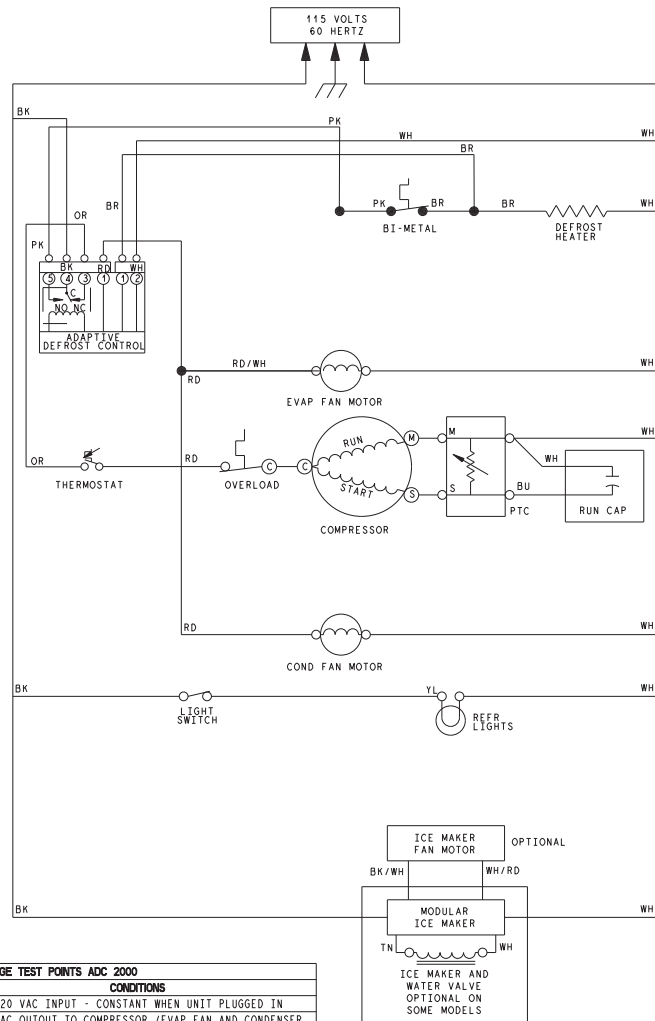
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WIRING DIAGRAM



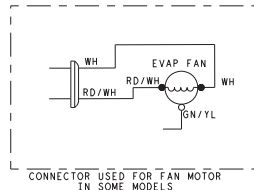
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VOLTAGE TEST POINTS ADC 2000					
MAIN CONTROL	FROM	COLOR	TO	COLOR	CONDITIONS
	P1	BK	P6	WH	120 VAC INPUT - CONSTANT WHEN UNIT PLUGGED IN
	P6	WH	P4	OR	120 VAC OUTOUT TO COMPRESSOR /EVAP FAN AND CONDENSER FAN WHEN COOLING
	P2	PK	P6	WH	120 VAC OUTPUT TO DEFROST HEATER WHEN ENERGIZED
	P1	BK	P5	BR	120 VAC INPUT DEFROST HEATER RUN-TIME FEEDBACK
	P1	BK	P3	RD	120 VAC INPUT COMPRESSOR RUN-TIME FEEDBACK

WIRE COLOR CODE

VI = VIOLET
BU = BLUE
BK = BLACK
RD = RED
WH = WHITE
YL = YELLOW
OR = ORANGE
BR = BROWN
GY = GRAY
PK = PINK
PU = PURPLE
TN = TAN
OR/BK = ORANGE/BLACK TRACER
YL/RD = YELLOW/RED TRACER
BU/BK = BLUE/BLACK TRACER
WH/BU = WHITE/BLUE TRACER
BK/YL = BLACK/YELLOW TRACER
WH/RD = WHITE/RED TRACER
GN/YL = GREEN/YELLOW TRACER
BK/WH = BLACK/WHITE TRACER
YL/BK = YELLOW/BLACK TRACER
BK/RD = BLACK/RED TRACER
RD/WH = RED/WHITE TRACER



DETAILS A

SYMBOL CODE

- ⊙ : CONNECTOR - SCREW ON
- ⊕ : CONNECTOR - CLOSED END
- : DISCONNECT TERMINAL
- : PERMANENT CONNECTION
- ➔ : PLUG CONNECTOR
- ⏏ : GROUND (CHASSIS)

MANUFACTURED UNDER ONE OR MORE
OF THE FOLLOWING UNITED STATES PATENTS

3,960,631 4,659,157 4,765,696 4,908,544 5,011,408
4,084,725 4,665,708 4,767,898 4,911,508 5,033,409
4,090,641 4,684,553 4,768,353 4,914,828 5,033,409
4,102,660 4,706,169 4,776,178 4,920,758 5,042,641
4,327,557 4,707,401 4,787,216 4,924,680 5,044,330
4,330,310 4,709,556 4,799,369 4,934,541 5,050,640
4,640,432 4,715,542 4,800,935 4,936,641 5,070,700
4,649,712 4,728,759 4,801,181 4,944,566 5,071,469
4,649,717 4,745,656 4,833,894 4,958,890 5,099,309
4,649,718 4,745,775 4,862,577 4,996,848

OTHER PATENTS PENDING

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